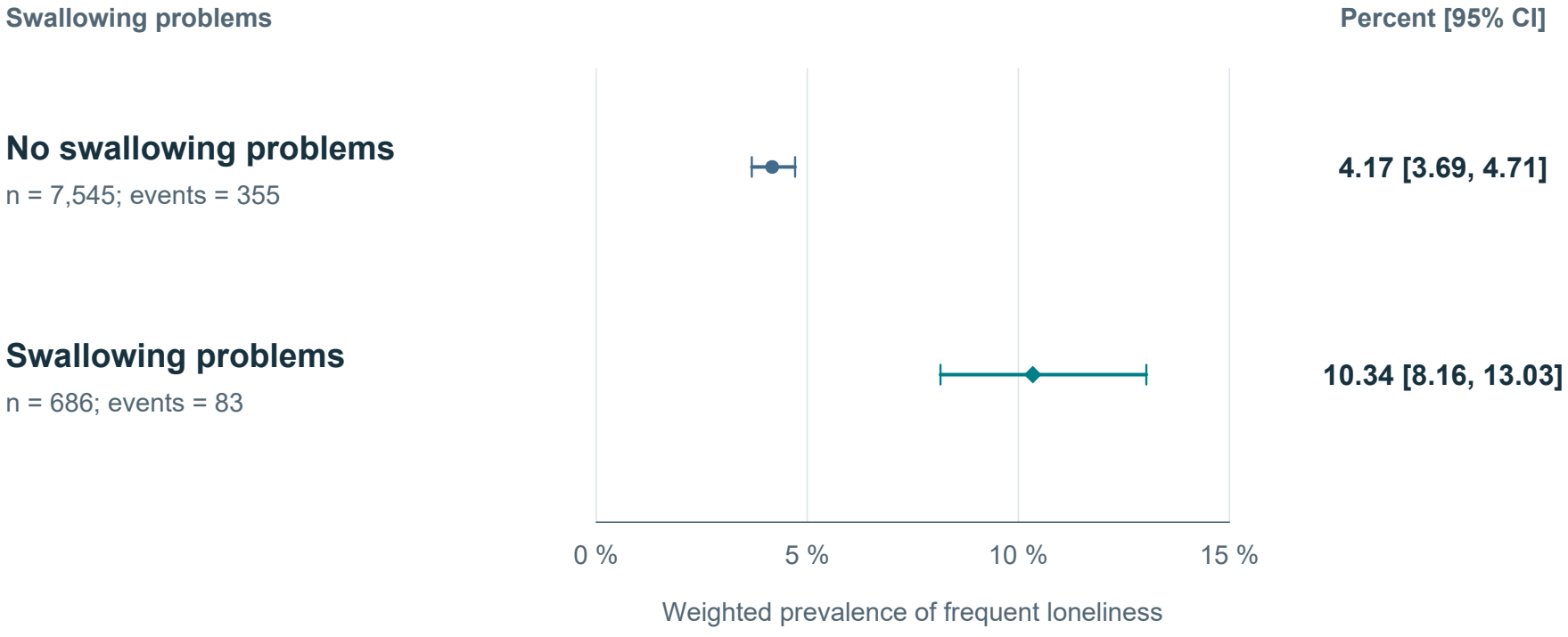
NHIS 2025 / FIGURE 1 / ADULTS AGED 65+

Frequent loneliness by swallowing problems

Available responses: N = 8,231 | unadjusted, survey-weighted estimates



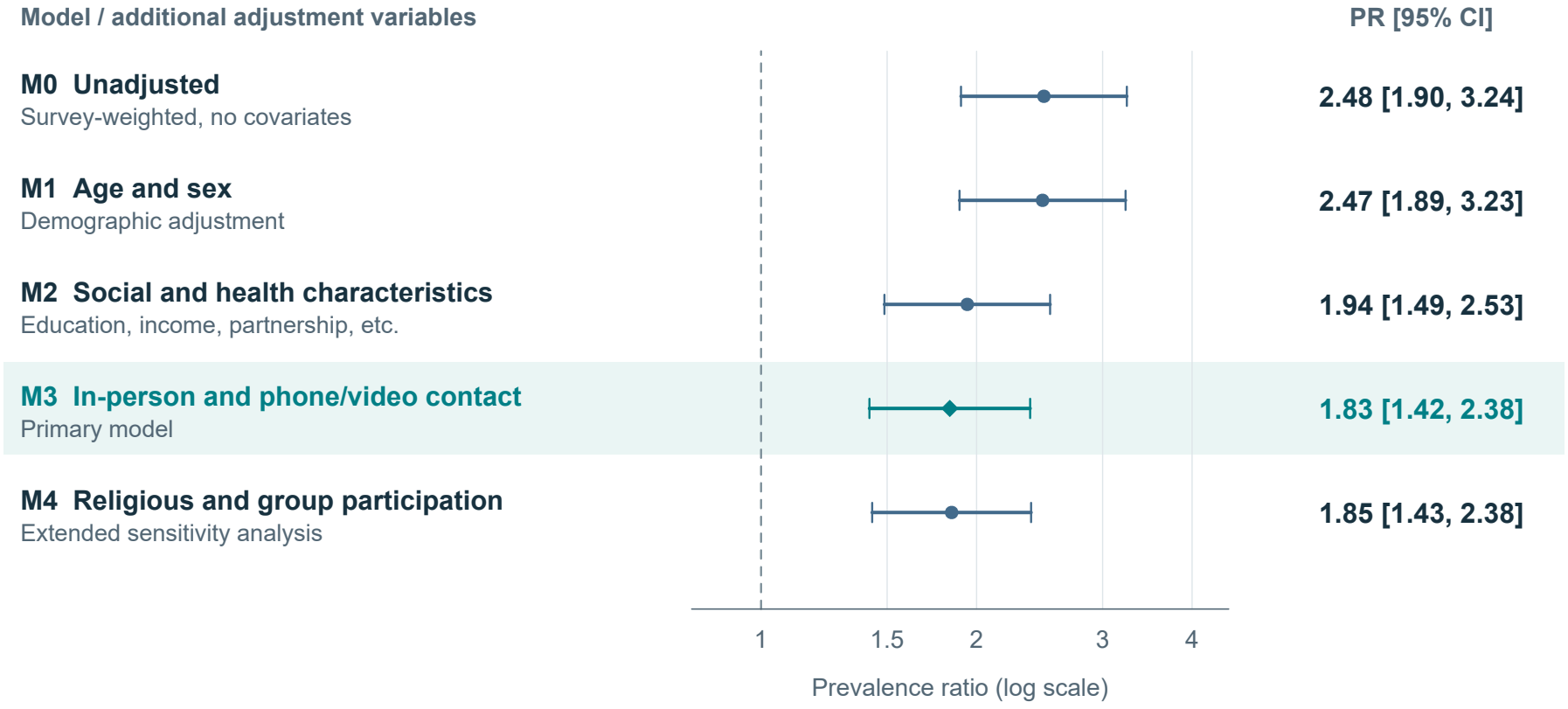
Frequent loneliness: always or usually lonely. Swallowing problems: self-reported during the past 12 months.
Points: weighted prevalence; lines: design-based 95% confidence intervals (CIs). Counts are unweighted.
No covariate adjustment. Denominators differ from the common regression sample.

Source: NHIS 2025 public-use data; survey-weighted analysis

NHIS 2025 / FIGURE 2 / ADULTS AGED 65+

Association after sequential adjustment

Frequent loneliness | swallowing problems vs. none | N = 8,094; 429 events



M2 adds education, family poverty ratio, partnership, living alone, mobility, hearing and history of stroke.
Models build on the preceding model. M3: contact outside the household; M4: participation in the past year.
Survey-weighted Poisson models; 95% CIs; ten income imputations. PR = 1 indicates no prevalence difference.
Frequent loneliness: always or usually lonely. Common sample; cross-sectional associations, not causal effects.

Source: NHIS 2025 public-use data; survey-weighted analysis

Association by severity of swallowing problems

Secondary analysis | M3 adjustment | N = 8,093; 429 events

Self-reported swallowing difficulties

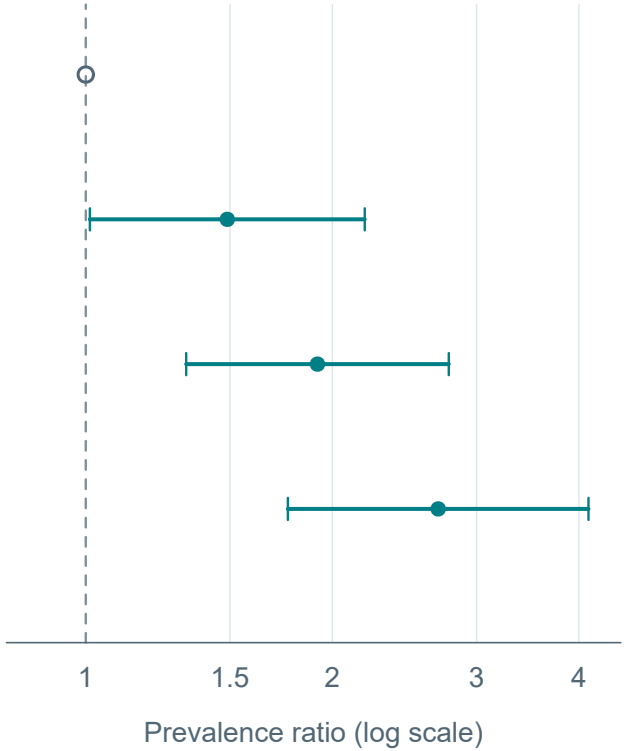
Adjusted PR [95% CI]

No swallowing problems
n = 7,420; events = 347
1.00 (reference)

A little difficulty
n = 361; events = 33
1.49 [1.01, 2.19]

Somewhere in between
n = 199; events = 27
1.92 [1.33, 2.77]

A lot of difficulty
n = 113; events = 22
2.69 [1.76, 4.11]



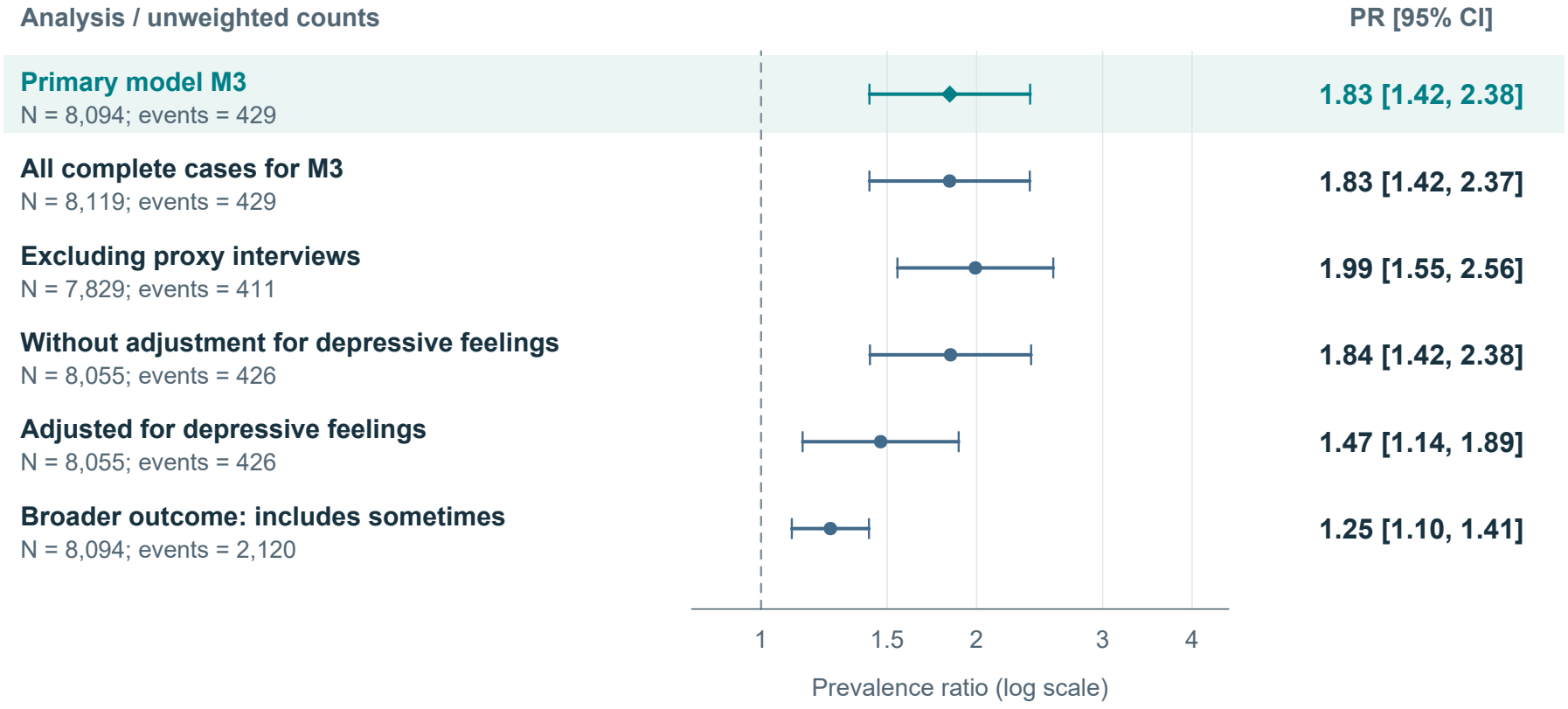
Frequent loneliness: always or usually lonely. Categories follow the codebook; severity is not a continuous scale.
M3: demographic, social and health characteristics plus in-person and phone/video contact.
Survey-weighted Poisson models; 95% CIs; ten income imputations. Sample sizes and event counts are unweighted.
Secondary results: only 22 events in the most severe category; no trend test or evidence of causation.

Source: NHIS 2025 public-use data; survey-weighted analysis

NHIS 2025 / SUPPLEMENTARY FIGURE S1 / ADULTS AGED 65+

Sensitivity analyses of the primary association

Prevalence ratios | sample composition or outcome definition varies across analyses



Matched comparison: identical sample (N = 8,055). Attenuation does not establish mediation.
Primary outcome: always or usually lonely; broader outcome: also sometimes. No multiplicity adjustment.
Models with depressive feelings or the broader outcome: some fitted values > 1; not individual probabilities.
Survey-weighted Poisson models; 95% CIs; ten income imputations. Odds-ratio sensitivity analysis not shown.

Source: NHIS 2025 public-use data; survey-weighted analysis